pushes the weaker to extinction, or each is driven into a narrower niche of specialism.

In 1976, however, American ecologist Hal Caswell proposed a “neutral” theory of biodiversity. It maintained that ecologically similar species are competitively equal, and whether species become common or rare is down to chance processes.

The “null” model

In the early 2000s, American ecologist Stephen P. Hubbell developed a mathematical model known as the “null” hypothesis, published in *The Unified Theory of Biodiversity and Geography* (2001), that supported Caswell’s theory. He tested his model by studying real communities.

Neutral theories of biodiversity have dominated community ecology in recent years. However, an Australian study of coral reefs, published in 2014, focusing on once-dominant species that have been almost lost to overfishing, did not support the theory. According to Hubbell, species are interchangeable, so others should have increased to take their place. The fact that this did not happen in this case suggests that the neutral theory is flawed. The question of what maintains diversity remains an open one.

“Caswell made a bold attempt to create a neutral theory of community organization.”

Stephen P. Hubbell

See also: Human activity and biodiversity • Island biogeography • Climax community • Open community theory

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